

Asef Hossain Khan

Dhaka, Bangladesh • +880 1878 120028 • asefhossain123@gmail.com
linkedin.com/in/asef • github.com/AsefHossainKhan • asefhossainkhan.github.io

Summary

Senior Software Engineer with nearly 5 years of experience building and shipping full-stack web and cloud applications for international clients at BJIT. Designs serverless AWS architectures with infrastructure as code (AWS CDK), builds React/Next.js frontends and Python/Node.js backends, and ships to production through CI/CD on AWS and Google Cloud. Experienced in building production AI and LLM systems with built-in observability. AI-first developer: uses AI coding agents (Claude Code, GitHub Copilot, Codex) and agentic workflows daily, and builds custom agent skills and MCP servers.

Skills

Languages: TypeScript, JavaScript, Python, Kotlin, SQL

Frontend: React, Next.js, responsive UI, WebSocket

Backend: Node.js, FastAPI, REST APIs, Socket.IO, microservices, serverless architecture, batch processing

Cloud & DevOps: AWS, AWS CDK (Infrastructure as Code), Google Cloud Run, Docker, CI/CD pipelines, LiveKit Cloud

AI & LLM: AI coding agents (Claude Code, GitHub Copilot, Codex, OpenCode), agentic development workflows, agent skills, MCP (Model Context Protocol) servers, LLM integration, prompt management, LiveKit Agents, Langfuse

Quality & Ops: Playwright (end-to-end testing), unit testing, code review, debugging, observability, AWS CloudWatch, custom metrics & dashboards

Data: MongoDB, SQL & NoSQL databases

Tools & Process: Git, GitHub, GitLab, Gerrit, Jira, Redmine, Agile

Mobile: Android (Kotlin, Jetpack Compose)

Experience

BJIT

Dhaka, Bangladesh

Senior Software Engineer

July 2025 – Present

- Own client projects end to end, from requirements and system design through production deployment and support
- Designed a healthcare client's full serverless AWS architecture and React (TypeScript) frontend from scratch, defined all infrastructure as code with AWS CDK (Python), and got client sign-off on the technology choices
- Lead teams of 2–3 engineers: break epics into tasks, review code, and onboard new team members
- Drive AI-assisted development with coding agents such as Claude Code and GitHub Copilot; built and shared custom agent skills adopted by colleagues, and built MCP servers that connect AI agents to internal systems and workflows
- Added LLM observability to a production voice agent with Langfuse, including custom metrics, dashboards, and user feedback tracking
- Write Playwright end-to-end tests for critical user flows and keep unit test suites passing before each release

BJIT

Dhaka, Bangladesh

Software Engineer

November 2021 – June 2025

- Developed full-stack web applications with React, Next.js, Node.js, and Python, as well as native Android apps (Kotlin, Java)
- Built REST APIs and real-time communication features using Socket.IO and WebSockets
- Set up CI/CD pipelines and shipped production deployments to AWS and Google Cloud Run using Docker containers
- Debugged production issues in serverless AWS workloads using CloudWatch logs and metrics
- Worked in agile teams: took part in code reviews, client requirement gathering, and technical solution proposals

- Supported a class of 40 students in a Data Structures course through tutoring, assignment reviews, and classroom activities

Projects

AI Voice Agent Platform:

- Designed and deployed a production AI voice agent system on LiveKit for real-time voice conversations
- Built the Next.js frontend and a FastAPI backend microservice for agent logic and conversation handling
- Deployed containerized services on Google Cloud Run with fully automated CI/CD across multiple instances, backed by MongoDB
- Integrated Langfuse for LLM observability, prompt management, custom metrics, and user feedback analytics
- Secured with dual-layer authentication using JWT and LiveKit access tokens

Health and Therapy Platform:

- Architected a fully serverless AWS platform, with infrastructure as code in AWS CDK (Python) and multiple React (TypeScript) web apps
- Built a batched serverless pipeline that processed 200+ Zoom session recordings through AI analysis on its first UAT run with zero pipeline failures
- Set up CI/CD pipelines for automated builds and deployments
- Designed secure handling of patient data in line with healthcare privacy standards, plus real-time patient–therapist communication

Road Safety Application:

- Built a mobile app for reporting traffic accidents, with a web dashboard for data analysis, as part of a World Bank open-source road safety project
- Ran user training sessions and fed the feedback into ongoing improvements

Education

Achievements

Employee of the Year: Awarded twice at BJIT (2022, 2026).

Valedictorian: Graduated as Valedictorian and Summa Cum Laude with a perfect 4.0 CGPA, 21st Convocation, AIUB.

Chancellor's Award: Presented by the then Minister of Education.

Media Recognition: Featured in [Prothom Alo](#) for outstanding academic achievements.

Training Excellence: Ranked 1st in the competitive BJIT Academy training program.

Dean's List: Consecutive semesters on the Dean's List (Fall 2018-2019, Spring 2018-2019).

Programming Competition: 3rd place in AIUB CS Fest 2018 Programming Contest.

1st in Dota 2: AIUB Cyber Gaming Fest 2019 (Team Game).